

Supervised Learning - Report

Kormákur Snorrason

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1 Introduction

The sport of mixed-martial arts (MMA) is widely regarded as one of the most unpredictable- and dynamic sports one can spectate. A sudden knockout, freak injury or a mistake giving way to a submission, it's a sport where anything, at any moment, can happen. When each bout is equally probable to be a controlled, one-sided domination as it has being an explosive war of violence and unpredictability, the question becomes: can an outcome of such an event even be predicted? Let alone consistently?

The UFC (Ultimate Fighting Championship) is currently the largest professional MMA organization with an extensive record of fights spanning across its 30+ years of existence. With such a long and lustrous history, finding a suitable dataset was not a difficult task and Kaggle, UFC Fight Data - 2026 provided just that.

With records dating back to March 2010 all the way to May of this year, it covers over 7000 fights and includes 57 features that describe each fighters stats., physical attributes and betting odds, and more. The set also includes a binary classifier on the winning fighter (whether the fighter in the red corner won or lost), which works as a great predictor feature.

Given this predictor feature we want to take note of the given baseline: the red fighter has a win rate of 58% against blue's 42%, which is a significant gap.

With this benchmark, the main goal of this project then becomes two-fold:

- Develop a model that can outperform the simple majority class baselines and accurately predict a UFC fight based purely on statistical differences between the fighters.
- See which attributes/features contribute the most to a outcome and by how much of a margin.

2 Process

Describe the process for finding the best model and justifying all decision. If you used other papers/sources for your decisions you need to reference them here.

After having picked a candidate dataset, we needed to verify its quality and usability for a potential model. For this part of the process we conducted some simple, but effective EDA that would answer those questions.

We first checked for any obvious outliers and/or missing data values through its official site, of which there were few and far between. Once that was done, we moved to briefly analysing the datasets features. What peaked our interest during this step was that the dataset didn't just include each fighters raw statistics, but instead it took the difference between each fighters raw statistical values, e.g., `sig_str_diff` (significant strikes difference between the red- and blue fighter). With the goal of the model being to predict a winner based on comparisons in such attributes, having them pre-calculated was a. Can We Predict UFC Fight Outcomes?

2.1 Data Cleaning

After having conducted some initial pruning into the dataset and verified its usability we proceeded to a more extensive data- cleaning and analysis for feature selection.

2.1.1 Outliers & Missing Data

The `_return` Columns

The values of the `red_return` and `blue_return` columns reflect the payout a \$1 bet would return. Including them would cause target leakage since they tell the model exactly who won (loser's return being 0) and in turn lead to a 100% model prediction rate accuracy; something we obviously wanted to avoid.

Stance Outliers

We noticed in the `red_stance` and `blue_stance` columns that there were a few rows with faulty classifications and missing data ("Switch " with a trailing space and "Unknown"), that needed to be excluded. We also found one other classification called "Open Stance", which describes an opposite stance matchup, which was only found in the `blue_stance` column. Since we already

have the three main stances ("Orthodox", "Southpaw" and "Switch") it becomes a redundant categorical value and only creates conflict.

Weight Class Outliers

In the column `weight_class` there we found the value "Catch Weight" which describes fights where one or both fighters missed their target weight and therefore fought outside one of the 12 regulatory weight classes. This is by definition an outlier to the standard procedure of an MMA bout and should therefore be excluded.

Outliers in `_diff` Columns

In the `rounds_diff` column we found one extreme outlier where the value was over 440. We excluded this due to it being an impossible data value when looking at either fighters records (i.e. total fights).

In the `reach_diff` column there is a single outlier where the reach difference was over 186 caused by one of the fighters reach being measured 0 cm. This is an obvious error caused by a missing data value so we naturally excluded it.

2.1.2 Feature Selection- & Engineering

Classifications

We decided to break down the dataset into the following classes:

Gender: MALE — FEMALE

Weight Classes:

Stance:

- Orthodox: Left foot forward, favored by right-handed fighters.
- Southpaw: Right foot forward, favored by left-handed fighters.
- Switch: Both Orthodox and Southpaw.

We initially included fighters age as classes, whose values were pooled into buckets of age ranges (e.g., 18 - 23) but a realization quickly arose. That is, as we increased the number of buckets (age ranges) the accuracy of the model increased in parallel, which suggest that there is too much variability in a fighters standing in even as little as a year. And since the dataset already included the age difference between fighters for each bout, which is the effect we're interested in, we decided to drop it entirely.

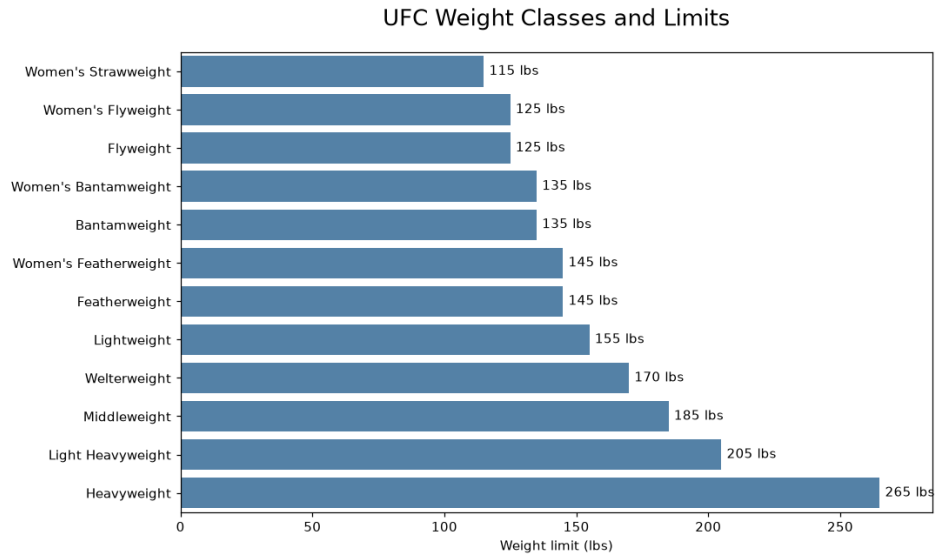


Figure 1: UFC Weight Classes

Feature Engineering

The impact of the age difference between two fighters varies significantly depending on where said difference occurs on the age-axis. In a bout between a 25 year old and a 28 year old, the age difference is not nearly as significant as one between a 32 year old and a 35 year old; and the data shows this. In just title fights, we see that fighters over 35 have a collective win-loss record of 50-75 across all weight classes:

However this distribution skews much more towards the heavier weight classes (Middleweight, Light Heavyweight and Heavyweight), which shows that the higher classes are a lot less forgiving on older fighters:

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Finding the right algorithm's for our models

some intro text

Logistic Regression

To establish a baseline for comparison, we will be comparing them to a simple logistic regression model.

Why did we choose logistic regression as our baseline for accuracy? Since our

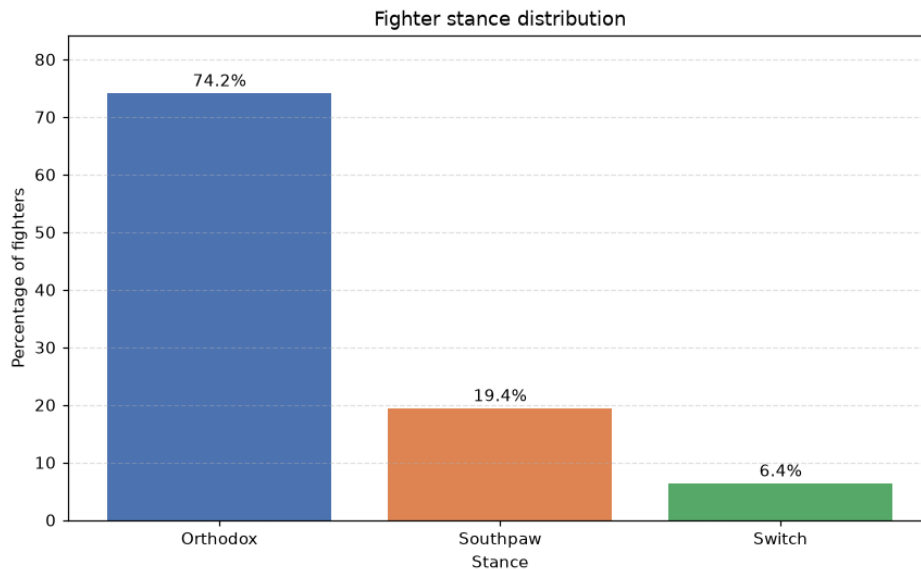


Figure 2: Fighters Stance Distribution

data is split between `_diff` features and other features, and the `_diff` features are numeric, that is what logistic regression is built for. So for example, each unit of reach advantage shifts the odds of red winning by some amount. Another reason for using logistic regression is it's fast to train and does not take a lot of setup.

A limitation of logistic regression is that it assumes a relatively simple relationship between the features and the log-odds of the outcome. It also does not automatically capture interactions between features. For example, maybe reach advantage has different effects based on what weight class the fight is in.

Random Forest

The reasons we chose random forest for one of our models are:

- Handles the mix of numeric `_diff` and categorical features without needing scaling, and works well with on-hot encoded categorical features.
- A major benefit of random forests is that they can capture non-linear relationships and interactions automatically, for example, the take-down advantage matters a lot more when the weight difference between the fighters is larger.

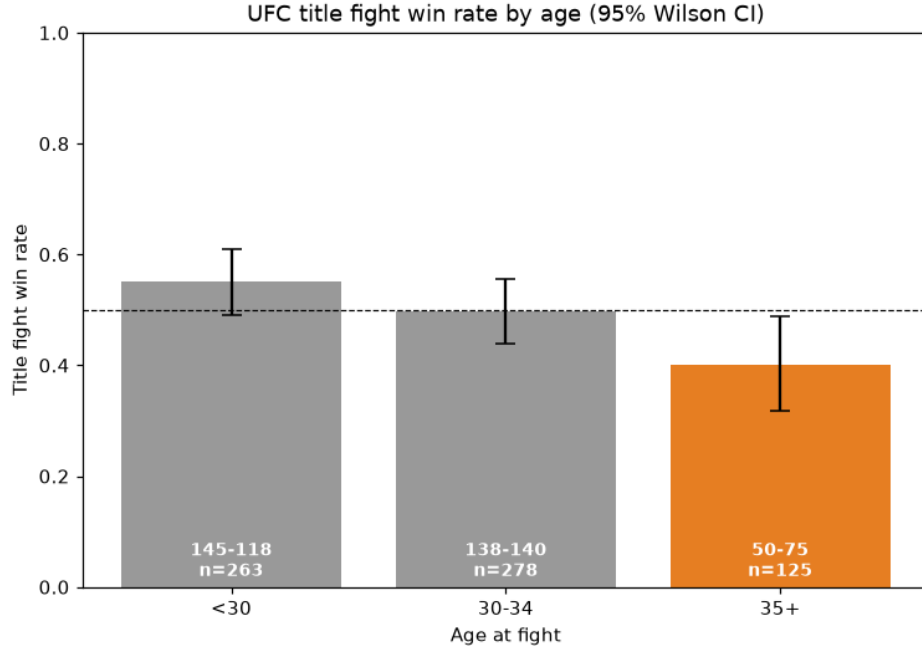


Figure 3: UFC title fight win rate by age (all weight classes)

XGBoost

The reasons we chose XGBoost for one of our models are:

- Like random forests, XGBoost can capture non-linear relationships and interactions. However, instead of averaging many independently trained trees, XGBoost builds trees sequentially, with each new tree attempting to correct errors made by the previous trees. Since our data set is not too big the training time for the models is not too long.
- Captures non-linear relationships and interactions like random forest, but often extracts more predictive power from the same features because each new tree learns from the previous ones.

Cons for XGBoost

- There are a lot of hyperparameters to tune, and if you don't tune them well the model can be prone to overfitting.
- Requires one-hot encoding for categorical features, which can cause the numbers of features to blow up quite quickly.

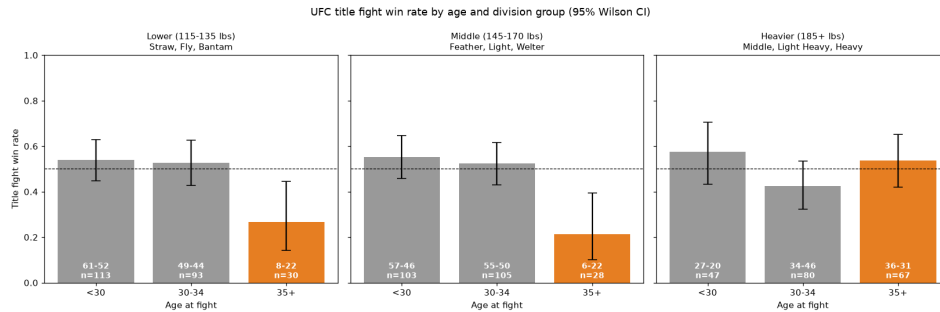


Figure 4: UFC title fight win rate by age & division

Why include it: XGBoost provides a useful comparison with random forest because both can capture non-linear relationships and interactions, but they use different ensemble approaches.

CatBoost

The reasons we chose CatBoost for one of our models are:

- Handles categorical features natively, instead of one-hot encoding, this is very useful since we have a lot of categorical features.
- CatBoost can make use of categorical features directly and can capture relationships between categorical variables, such as red_{stance} and $blue_{stance}$. This has a big impact on

Cons for CatBoost

- Is quite similar to XGBoost. It can be slower to train than some of the other models, particularly when using more complex settings.

Why include it: this gives a good way to test whether the native category handling matters, and a good way to find that out is compare XGBoost vs CatBoost.

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3 Results

Report on the performance for the different models and hyperparameters. Essentially, take the numbers you got as output of your program and put them into a nice form (plots, tables, ...) that makes them easier to interpret and shows, for example, how the choice of values for different hyperparameters influences the performance.

4 Conclusion

Interpret the results and compare with results in the literature (if possible). What can be said about the performance of different models on the problem? How important is the choice of the hyperparameters for the performance of the different models?

5 Future Work

Suggest how your results could be improved. What did you learn from the results you got and what would you do differently now that you have learned this?